

ADVANCED AND QUANTUM OPTICS — Paper 2

Model Solutions and Marking Scheme

Natural Sciences Tripos Part II — 2026

SECTION A

Question 1 [4 marks] — Coherence of a Gaussian Source

For a Gaussian spectral profile with FWHM $\Delta\nu = 1.5$ GHz, the coherence time is

$$\tau_c = \frac{\sqrt{2 \ln 2 / \pi}}{\Delta\nu} = \frac{\sqrt{0.4413}}{1.5 \times 10^9} = \frac{0.664}{1.5 \times 10^9} = 4.43 \times 10^{-10} \text{ s} = 443 \text{ ps}.$$

[1 mark]

The coherence length is

$$l_c = c \tau_c = 3 \times 10^8 \times 4.43 \times 10^{-10} = 0.133 \text{ m} = 13.3 \text{ cm}.$$

[1 mark]

For a Gaussian profile, $|g(\tau)| = \exp(-\pi\tau^2/(2\tau_c^2))$ in the power-equivalent-width convention, but more directly using the standard result:

$$|g(\tau)| = \exp\left(-\frac{\pi(\Delta\nu)^2\tau^2}{4 \ln 2}\right).$$

At $\tau = 200 \text{ ps} = 2.0 \times 10^{-10} \text{ s}$:

$$\frac{\pi(\Delta\nu)^2\tau^2}{4 \ln 2} = \frac{\pi(1.5 \times 10^9)^2(2.0 \times 10^{-10})^2}{4 \times 0.6931} = \frac{\pi \times 0.09}{2.773} = 0.1019.$$

$$|g(\tau)| = e^{-0.102} = 0.903.$$

[2 marks] for correct substitution and final value.

Question 2 [4 marks] — 4f Imaging System

The total system is: propagation f_1 – thin lens f_1 – propagation $f_1 + f_2$ – thin lens f_2 – propagation f_2 . The overall matrix is

$$M = M_{d_4} M_{L_2} M_{d_{23}} M_{L_1} M_{d_1},$$

where $d_1 = f_1$, $d_{23} = f_1 + f_2$, $d_4 = f_2$.

Step 1: Propagation f_1 then lens f_1 :

$$M_1 = \begin{pmatrix} 1 & 0 \\ -1/f_1 & 1 \end{pmatrix} \begin{pmatrix} 1 & f_1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & f_1 \\ -1/f_1 & 0 \end{pmatrix}.$$

Step 2: Propagation $f_1 + f_2$:

$$M_2 = \begin{pmatrix} 1 & f_1 + f_2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & f_1 \\ -1/f_1 & 0 \end{pmatrix} = \begin{pmatrix} 1 - (f_1 + f_2)/f_1 & f_1 \\ -1/f_1 & 0 \end{pmatrix} = \begin{pmatrix} -f_2/f_1 & f_1 \\ -1/f_1 & 0 \end{pmatrix}.$$

Step 3: Lens f_2 :

$$M_3 = \begin{pmatrix} 1 & 0 \\ -1/f_2 & 1 \end{pmatrix} \begin{pmatrix} -f_2/f_1 & f_1 \\ -1/f_1 & 0 \end{pmatrix} = \begin{pmatrix} -f_2/f_1 & f_1 \\ 0 & -f_1/f_2 \end{pmatrix}.$$

Step 4: Final propagation f_2 :

$$M = \begin{pmatrix} 1 & f_2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -f_2/f_1 & f_1 \\ 0 & -f_1/f_2 \end{pmatrix} = \begin{pmatrix} -f_2/f_1 & 0 \\ 0 & -f_1/f_2 \end{pmatrix}.$$

[2 marks] for correct matrix multiplication (can quote the diagonal result directly for $4f$ systems if justified).

The matrix is diagonal, confirming that the system is a perfect imaging system (the output ray position depends only on the input position, not the input angle). The lateral magnification is

$$m = -\frac{f_2}{f_1} = -\frac{20}{10} = -2.$$

The image is inverted and magnified by a factor of 2. **[2 marks]**

Question 3 [4 marks] — Acousto-Optic Modulator

The acoustic wavelength is

$$\Lambda = \frac{v_s}{f} = \frac{6 \times 10^3}{50 \times 10^6} = 1.2 \times 10^{-4} \text{ m} = 120 \text{ } \mu\text{m}.$$

[1 mark]

The Bragg angle (measured from the acoustic wavefronts, i.e. from the direction perpendicular to the sound wave) is given by the Bragg condition $2\Lambda \sin \theta_B = \lambda$, where $\lambda = \lambda_0/n$:

$$\sin \theta_B = \frac{\lambda_0}{2n\Lambda} = \frac{633 \times 10^{-9}}{2 \times 1.46 \times 1.2 \times 10^{-4}} = \frac{633 \times 10^{-9}}{3.504 \times 10^{-4}} = 1.81 \times 10^{-3}.$$

Since θ_B is small, $\theta_B \approx 1.81 \text{ mrad} = 0.104^\circ$. **[2 marks]**

The first-order diffracted beam is frequency-shifted by exactly the acoustic frequency:

$$\Delta\nu = f = 50 \text{ MHz}.$$

(Upshifted for the +1 order, downshifted for −1 order.) **[1 mark]**

Question 4 [4 marks] — **Optical Fibre Parameters**

The numerical aperture is

$$\text{NA} = \sqrt{n_1^2 - n_2^2} = \sqrt{1.462^2 - 1.457^2} = \sqrt{2.1374 - 2.1228} = \sqrt{0.01459} = 0.1208.$$

[1 mark]

The critical angle for total internal reflection at the core-cladding interface is

$$\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right) = \sin^{-1}\left(\frac{1.457}{1.462}\right) = \sin^{-1}(0.99658) = 85.3^\circ.$$

The complement (maximum ray angle from the axis) is $90^\circ - 85.3^\circ = 4.7^\circ$, consistent with the NA. [1 mark]

For single-mode operation, we require the normalised frequency $V < 2.405$:

$$V = \frac{\pi d_{\text{core}}}{\lambda_0} \text{NA} = \frac{\pi \times 8.0 \times 10^{-6}}{1.55 \times 10^{-6}} \times 0.1208 = 16.20 \times 0.1208 = 1.96.$$

Since $V = 1.96 < 2.405$, the fibre is **single-mode** at $\lambda_0 = 1.55 \mu\text{m}$. [2 marks]

SECTION B

Question 5 [17 marks] — Resonators and Laser Oscillation

(a) [5 marks]

An initial wave U_0 enters the resonator and undergoes multiple round trips. After each round trip the wave acquires a phase $\phi = k \cdot 2d$ and is attenuated by the complex factor $h = |r|e^{-j\phi}$, where $|r|^2$ is the round-trip intensity attenuation. The total intracavity field is the geometric series:

$$U = U_0 + hU_0 + h^2U_0 + \cdots = \frac{U_0}{1 - h}.$$

[1 mark] for the geometric series.

The intracavity intensity is

$$I = |U|^2 = \frac{|U_0|^2}{|1 - h|^2} = \frac{I_0}{|1 - |r|e^{-j\phi}|^2}.$$

Expanding the denominator:

$$|1 - |r|e^{-j\phi}|^2 = 1 + |r|^2 - 2|r|\cos\phi.$$

[1 mark]

Using the identity $1 - \cos\phi = 2\sin^2(\phi/2)$:

$$1 + |r|^2 - 2|r|\cos\phi = (1 - |r|)^2 + 4|r|\sin^2(\phi/2).$$

[1 mark]

Therefore

$$I = \frac{I_0/(1 - |r|)^2}{1 + \frac{4|r|}{(1 - |r|)^2} \sin^2(\phi/2)} = \frac{I_{\max}}{1 + (2\mathcal{F}/\pi)^2 \sin^2(\phi/2)},$$

where

$$I_{\max} = \frac{I_0}{(1 - |r|)^2}, \quad \mathcal{F} = \frac{\pi\sqrt{|r|}}{1 - |r|}.$$

[2 marks] for final expression and correct identification of finesse.

(b) [3 marks]

Resonances occur when $\phi = q \cdot 2\pi$, i.e. $k \cdot 2d = q \cdot 2\pi$:

$$\nu_q = q \frac{c}{2nd}, \quad q = 1, 2, 3, \dots$$

[1 mark]

The free spectral range (spacing between adjacent modes) is

$$\nu_F = \frac{c}{2nd}.$$

[1 mark]

The FWHM spectral width of each mode is

$$\delta\nu = \frac{\nu_F}{\mathcal{F}}.$$

[1 mark]

(c) [4 marks]

Stability condition: A spherical-mirror resonator is stable if and only if

$$0 \leq g_1 g_2 \leq 1, \quad \text{where } g_i = 1 - d/R_i.$$

[1 mark]

Stability diagram: In the (g_1, g_2) plane, the stable region is bounded by the hyperbola $g_1 g_2 = 1$ and the axes $g_1 g_2 = 0$. [1 mark for sketch]

Special configurations:

- **Planar:** $g_1 = g_2 = 1$ ($R_1 = R_2 = \infty$) — corner of stable region (marginally stable).
- **Confocal:** $g_1 = g_2 = 0$ ($R_1 = R_2 = d$) — origin.
- **Concentric:** $g_1 = g_2 = -1$ ($R_1 = R_2 = d/2$) — opposite corner (marginally stable).
- **Symmetric:** $g_1 = g_2$, lying along the diagonal.

[2 marks] for correctly placed special configurations. (Concentric and planar lie on the boundary $g_1 g_2 = 1$.)

(d) [5 marks]

Free spectral range:

$$\nu_F = \frac{c}{2d} = \frac{3 \times 10^8}{2 \times 0.15} = 1.00 \text{ GHz}.$$

[0.5 marks]

Number of longitudinal modes within the gain bandwidth:

$$M \approx \frac{\Delta\nu_D}{\nu_F} = \frac{300 \times 10^9}{1.00 \times 10^9} = 300 \text{ modes}.$$

[1 mark]

Loss coefficient: With $\mathcal{R}_1 = 1.00$ and $\mathcal{R}_2 = 0.95$, and $\alpha_s = 0$:

$$\alpha_r = \frac{1}{2d} \ln \frac{1}{\mathcal{R}_1 \mathcal{R}_2} = \frac{1}{0.30} \ln \frac{1}{0.95} = \frac{0.05129}{0.30} = 0.171 \text{ m}^{-1}.$$

[1 mark]

Photon lifetime:

$$\tau_p = \frac{1}{\alpha_r c} = \frac{1}{0.171 \times 3 \times 10^8} = 19.5 \text{ ns}.$$

[1 mark]

Threshold check: The laser reaches threshold if $\gamma_0 > \alpha_r$. Here $\gamma_0 = 0.20 \text{ m}^{-1} > \alpha_r = 0.171 \text{ m}^{-1}$. The gain exceeds the loss, so **the laser does reach threshold**. The ratio $\gamma_0/\alpha_r = 1.17$, so the laser operates about 17% above threshold. [1.5 marks]

Question 6 [17 marks] — Nonlinear Optics and Electro-Optic Effects

(a) [4 marks]

In the anharmonic oscillator model, the potential is $V = \frac{1}{2}k_0x^2 + \frac{1}{3}k_1x^3$. The restoring force is $F = -k_0x - k_1x^2$, giving a nonlinear response. [0.5 marks]

For a monochromatic driving field $E(t) = E_0 \cos(\omega t)$, the displacement x contains a linear term $\propto \cos(\omega t)$ plus a correction $\propto E^2 \propto \cos^2(\omega t)$. Using $\cos^2(\omega t) = \frac{1}{2}[1 + \cos(2\omega t)]$, the polarisation contains: [1 mark]

- A component at frequency 2ω — **second-harmonic generation**.
- A DC (zero-frequency) component — **optical rectification**.

[1 mark] for identifying both terms with physical names.

Centrosymmetric media: If the medium has inversion symmetry, the potential must satisfy $V(x) = V(-x)$. The cubic term $\frac{1}{3}k_1x^3$ changes sign under $x \rightarrow -x$ and therefore must vanish: $k_1 = 0$. Equivalently, $P(-E) = -P(E)$ in a centrosymmetric medium, which forbids even powers of E in the polarisation, ruling out all $\chi^{(2)}$ effects. [1.5 marks]

(b) [4 marks]

With $E(t) = \text{Re}[E(\omega_1)e^{i\omega_1 t} + E(\omega_2)e^{i\omega_2 t}]$, the second-order nonlinear polarisation $P^{(2)} \propto E^2$ generates five distinct frequency components:

1. $P^{(2)}(2\omega_1) = d E(\omega_1)E(\omega_1)$ — **second-harmonic generation** of ω_1 .
2. $P^{(2)}(2\omega_2) = d E(\omega_2)E(\omega_2)$ — **second-harmonic generation** of ω_2 .
3. $P^{(2)}(\omega_1 + \omega_2) = 2d E(\omega_1)E(\omega_2)$ — **sum-frequency generation**.
4. $P^{(2)}(\omega_1 - \omega_2) = 2d E(\omega_1)E^*(\omega_2)$ — **difference-frequency generation**.
5. $P^{(2)}(0) = d(|E(\omega_1)|^2 + |E(\omega_2)|^2)$ — **optical rectification**.

[1 mark] for listing all five frequencies correctly. [1 mark] each for correctly identifying at least three processes. [1 mark] for correct amplitude expressions.

(c) [5 marks]

Optical Kerr effect: In a medium with a third-order nonlinearity, the refractive index depends on intensity:

$$n(I) = n_0 + n_2 I, \quad \text{where} \quad n_2 = \frac{\chi^{(3)}(\omega; \omega, \omega, -\omega)}{2n_0^2 \varepsilon_0 c}.$$

n_2 has units of cm^2/W . For most materials $n_2 > 0$ (self-focusing). [1 mark]

Self-phase modulation: For a pulse $I(t)$ propagating through a Kerr medium of length z , the phase is

$$\varphi(t) = \omega_0 t - k_0 z [n_0 + n_2 I(t)].$$

The instantaneous frequency is

$$\omega(t) = \frac{\partial \varphi}{\partial t} = \omega_0 - k_0 z n_2 \frac{\partial I}{\partial t}.$$

[1.5 marks]

On the leading edge of the pulse, $\partial I/\partial t > 0$, so $\omega(t) < \omega_0$ (red-shifted). On the trailing edge, $\partial I/\partial t < 0$, so $\omega(t) > \omega_0$ (blue-shifted). This produces an approximately linear frequency chirp across the pulse centre, with lower frequencies at the front and higher frequencies at the back. [1.5 marks]

Compression vs. broadening: Whether this chirp leads to pulse compression or broadening depends on the sign of the group-velocity dispersion (GVD). In a medium with *anomalous* (negative) GVD, higher frequencies travel faster, so the blue-shifted trailing edge catches up with the red-shifted leading edge, leading to compression. In *normal* (positive) GVD, the chirp exacerbates the dispersive spreading and the pulse broadens. [1 mark]

(d) [4 marks]

Soliton formation: In a medium with anomalous GVD ($\partial^2 k/\partial \omega^2 < 0$), dispersion alone causes a pulse to broaden because different frequency components travel at different speeds. The Kerr effect ($n_2 > 0$) causes self-phase modulation, which generates new frequencies that — in the anomalous dispersion regime — counteract the dispersive broadening. When these two effects exactly balance, the pulse propagates without changing shape: this is a *temporal soliton*. [1.5 marks]

Nonlinear Schrödinger equation: The pulse envelope $A(x, z)$ satisfies

$$\frac{\partial^2 A}{\partial x^2} + \frac{n^2 n_2}{\eta_0} k_0^2 |A|^2 A = j2k \frac{\partial A}{\partial z},$$

where $\eta_0 = \sqrt{\mu_0/\epsilon_0}$ is the impedance of free space. (In the temporal analogue, x is replaced by the retarded time coordinate and the second derivative represents GVD.) [1.5 marks]

Soliton solution:

$$A(x, z) = A_0 \operatorname{sech}\left(\frac{x}{W_0}\right) \exp\left(-j\frac{z}{4z_0}\right),$$

where W_0 is the soliton width and $z_0 = \frac{1}{2}kW_0^2$ is the associated diffraction/dispersion length. The sech profile maintains its shape indefinitely. [1 mark]